

k h n t m i t n G E T t i Y e i
a t i o p i E o e y y h y k
h t n t i G y a o a E y o E i y
o t p h y a a t G k a p p a
p i e a y p h G n G a a n t e h
p p p y h i o p o p y E o i
G t k p i n E a n
E k k t
E i k

Google Sans Build Process

Project Plan Presentation

07 December 2020



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Project Goals

Project Goals

Main goal: actually merge the fonts

- Our proposed plan focuses on this
- From the 200-hour budget:
 - 63h to migrate the existing project to UFOs
 - 73h to define the merge process, document, align with vendors
 - 63 remaining hours split between 10 scripts = 6 hours to actually merge each vendor's delivery
- We would actually need 30h per script, more for scripts that require complex engineering such as Devanagari
- Total estimated plan is about 590h

Stretch goals: improve tooling

- Extend the role of CI (GitHub Actions)
- Add automated testing (e.g. Font Bakery)
- Glyphs.app 3 format full support in fontmake
- Glyphs.app Smart features support

Options

Option A

Merged UFOs, we handle the merges

- Source format: single set of UFOs + designspace
 - All scripts mixed in the same set of UFOs
- Dalton Maag receives .glyphs files or UFOs from vendors
- Dalton Maag merges all the scripts together

This is our recommended option:

- most supported by existing tooling: fontmake, glyphsLib
- most efficient: makes the best use of our expertise
- least adaptation from vendors

Option B

Merged UFOs, we train vendors to use git

- Source format: single set of UFOs + designspace
 - All scripts mixed in the same set of UFOs
- Dalton Maag trains vendors to use git to merge their own work
- Dalton Maag supervises the merges (review PRs)

We're discussing this option because Chris suggested it during our first meeting, however we can't recommend it:

- least efficient
 - training first and then reviewing big merge requests is probably ten times the amount of work compared to handling the merge ourselves
- lots of adaptation needed from vendors

Option C

Per-script UFOs, Merge on the Fly

- Source format: per-script sets of UFOs + designspace
 - for example, one folder per script, with independent sets of UFOs
 - Dalton Maag receives .glyphs files or UFOs from vendors
 - we place each vendor's delivery in their own folder, no merging at the source level
 - Dalton Maag maintains an automated merged build pipeline
 - each script compiles into its own set of TTFs
 - the TTFs are merged on the file into a single font file
- Mentioned but not recommended either:
- not supported by tooling
 - pyftmerge exists but isn't good enough ([fonttools#1696](#))
 - not our expertise

Project Plan

Project Plan

See spreadsheet:

[Google Sans Build Process - Project Plan](#)

Phases:

- 01 Move existing work to UFO format
- 02 Add export/import script
- 03 Document merge process, add assistive scripts
- 04 Merge work from vendors

Extras:

- 01 Investigate potential file size savings
- 02 Help augment the CI and add tests
- 03 Improve export/import script

Phase 1

Migrate to UFO

Exploratory work [here](#)

- 01 Turn the existing Glyphs files into UFOs+designspace
 - Make sure no information is lost in the process
 - Make sure the fonts built from UFOs are identical
 - Then delete the Glyphs files, declare UFOs the source of truth
- 02 Document existing informal best practices [to be reviewed by Chris]
 - Normalization
 - Feature file style guide
 - Kerning class naming
 - ...
- 03 Apply best practices to the UFOs+designspace

Phase 1 Notes

We want to document existing best practices on the repository, that Chris has been enforcing so far, and add a few (example: kerning class naming to allow smooth merging of all the scripts).

Chris will review our added guidelines, check that they won't take too much effort for vendors to comply, and share them with vendors.

Phase 2

Add an Export/Import Script

Idea: allow vendors to work on the source files using Glyphs.app

- 01 Create a script that exports UFOs into a Glyphs file and back
 - Ensure that no information is lost in the process
 - Deal with Glyphs.app 3
- 02 Adapt the CI
 - Generate the Glyphs files automatically from the latest UFOs
 - Still build fonts from vendor deliveries when pushed to a branch, as it is now

- 03 Document how to use the script, for vendors
 - They start by running the export script to get a Glyphs.app file from the latest sources
 - Then work in Glyphs.app
 - Then either keep working in Glyphs.app until the end, and deliver the Glyphs file for merging
 - Or turn their Glyphs file back to UFO, commit to git and deliver a branch to merge

Phase 2 Notes

We assume vendors will work in:

- UFO-based tools
 - can work directly on UFOs
- Glyphs.app version 2
 - existing tooling in glyphsLib
- Glyphs.app version 3
 - our planning includes time to look for a workaround
 - if not possible we might have to:
 - enforce Glyphs.app version 2 deliveries
 - or work on support for Glyphs.app 3 in glyphsLib

Phase 3

Document Merge Process + Scripts

Beyond technicalities of file formats, ensure smooth integration of the vendors' work in terms of scope, engineering, interaction with other scripts.

- 01 Document our merge process to set expectations from vendors
 - Each vendor will have a scope defined by which glyphs they should design, which features they should write, what they should kern, etc.
 - Each delivery will need to meet technical expectations in terms of best practices

- Our merge process will take the vendor's delivery for anything within the scope, and stick with what was already in the Google Sans repository for anything outside

- 02 Write scripts to help us through the work, if needed
 - e.g. a UFO merging script that cherry-picks glyph data from a list of glyph names

Phase 3 Notes

Chris will communicate our process and the project's requirements to vendors, and will make sure that files that we receive are meeting them.

Chris will also handle on a weekly basis potential out-of-scope fixes, so that if one vendor finds issues with the script of another, these issues get fixed by the appropriate vendor.

Phase 4

Merge Work from Vendors

This phase will happen for each delivery from the vendors:

- we suggest to merge work in progress a first time
 - as a proof-of-concept and to flag issues early before the final delivery
- then a second time for the final delivery

- 01 Merge expected changes using a script that overwrites the target for everything within the vendor's remit
- 02 Merge by hand things that overlap with existing work, or that cannot be automated (e.g. kerning, feature files)
- 03 QA the result of the merge

Phase 4 Notes

Chris will be our communication point in case we need more info from a vendor to perform a merge.

Extra 1

Investigate potential file size savings

- The current GPOS is about 200kb, including:
 - about 20kb due to anchor propagation
 - about 180kb due to kerning

Anchors are currently propagated in every glyph that is eligible. This might not reflect the designers' intention, and restricting anchor propagation to only the glyphs that are meant to work as such could save some bytes.

The current kerning visibly contracts the text when turned on; this could indicate that it is suboptimal and that it could partially be replaced by spacing, potentially leading to size savings.

Extra 2

Help Augment the CI and Add Tests

Under Chris's supervision, we can assist in:

- adding more capabilities to the CI
 - e.g. automated regression testing
- growing the project's test suite
 - e.g. custom Font Bakery profile

Extra 3

Improve the Export/Import Script

Add support for:

- Glyphs 3 file format
- smart components
- smart corners

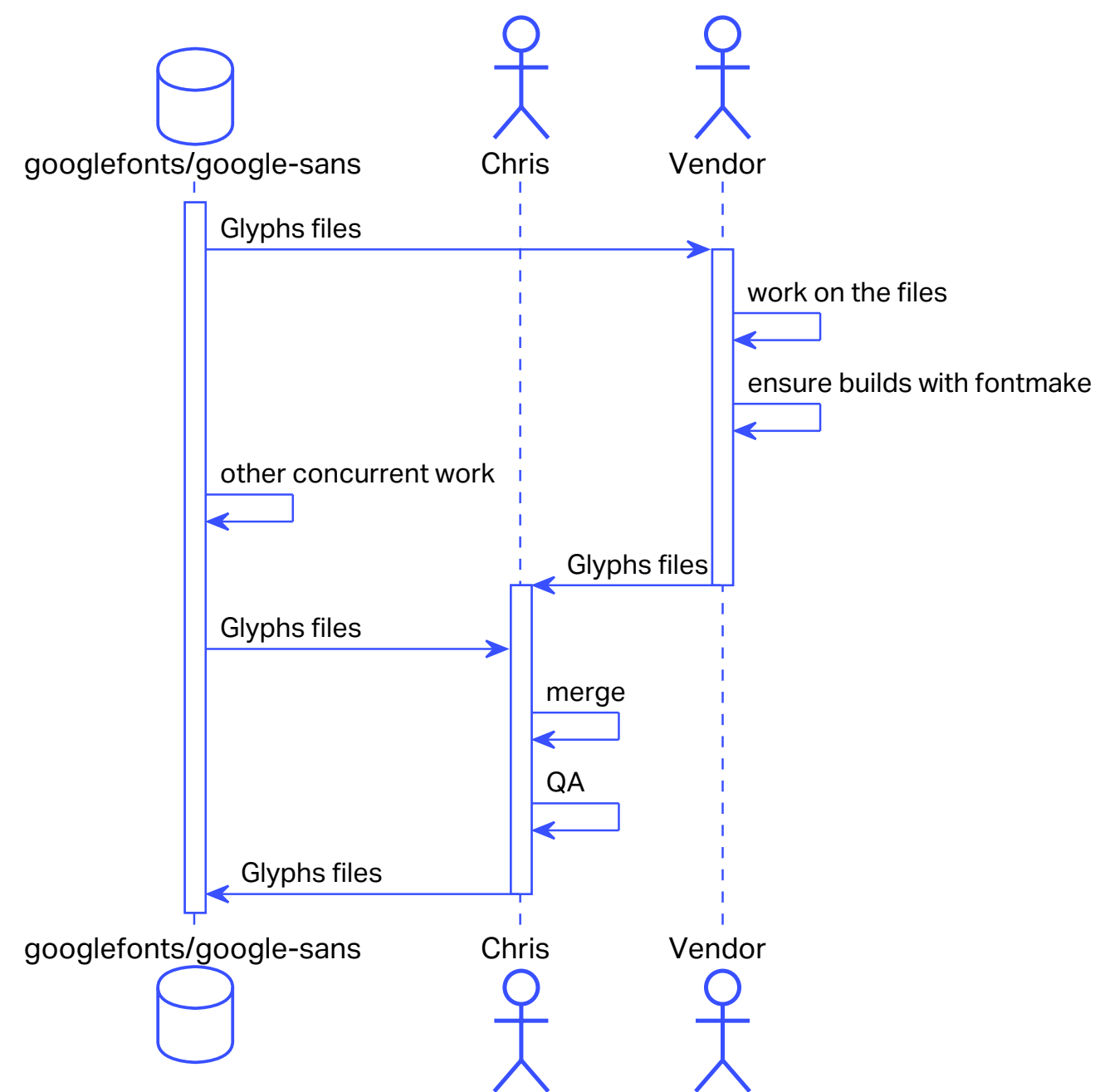
Detailed Merge Process

Overview

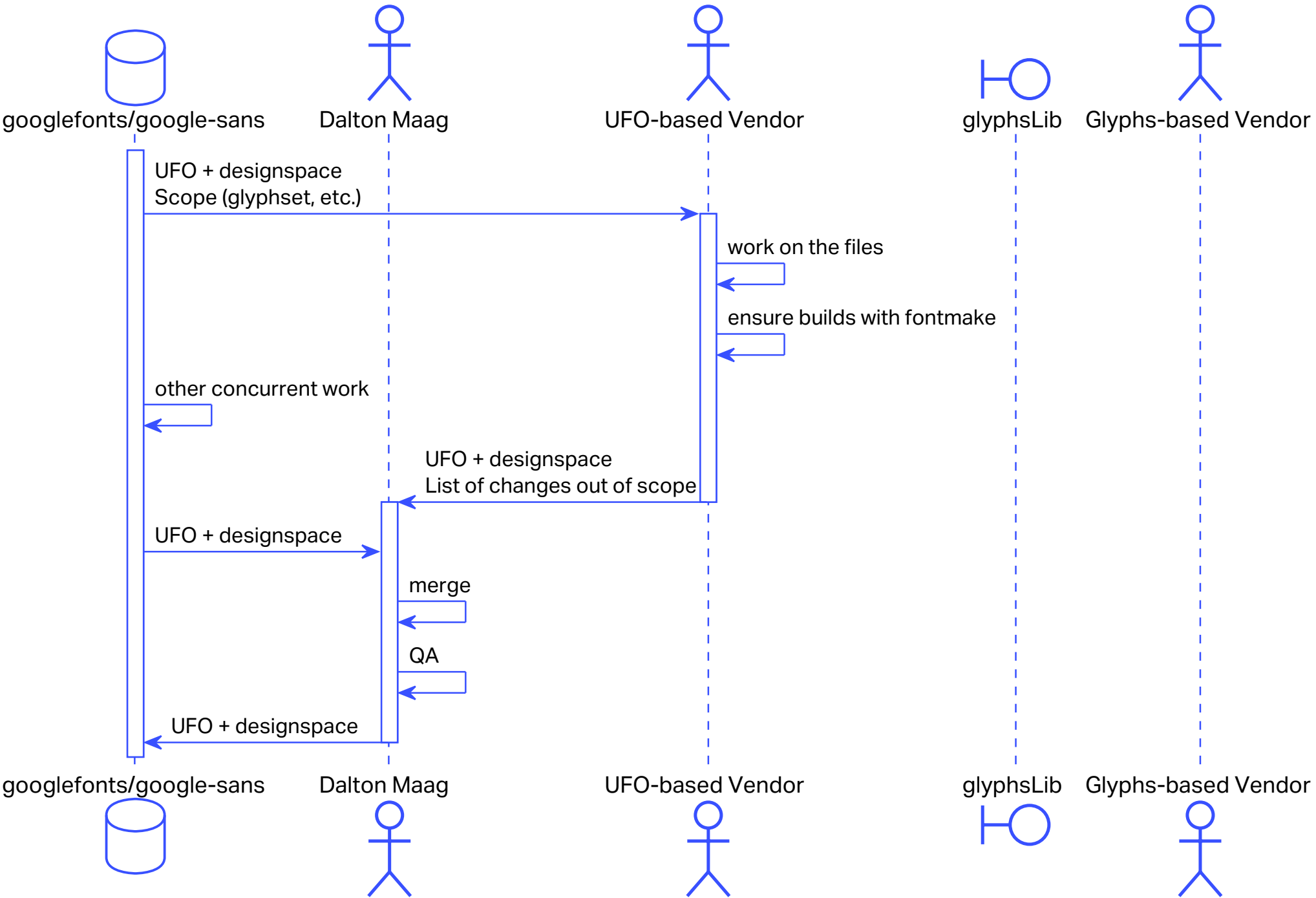
Key points

- vendors work either on UFOs or on Glyphs files
- Dalton Maag merges at the UFO level

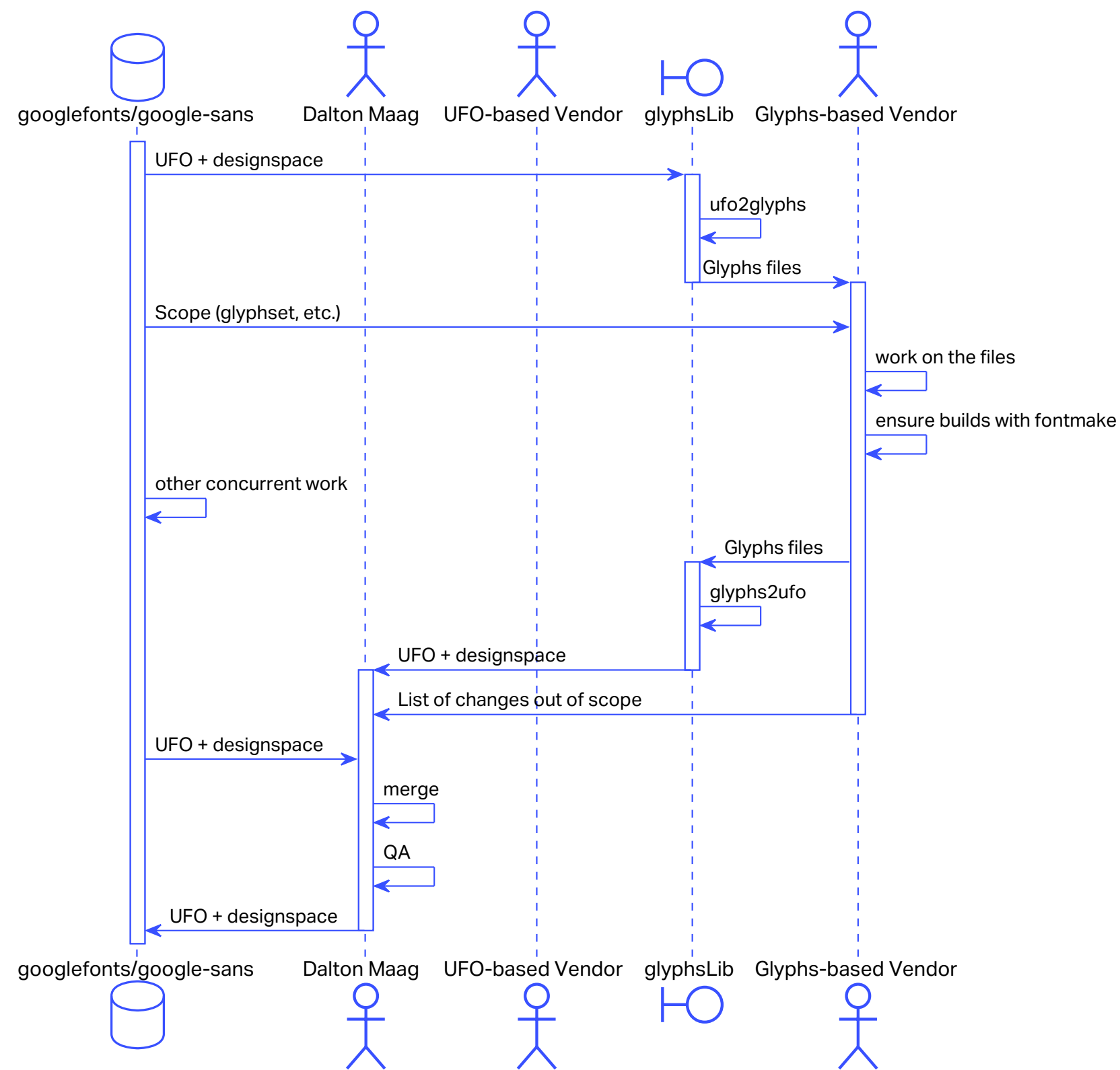
Google Sans Build Process – Current Merge Process



Google Sans Build Process – New Merge Process – UFO Vendor



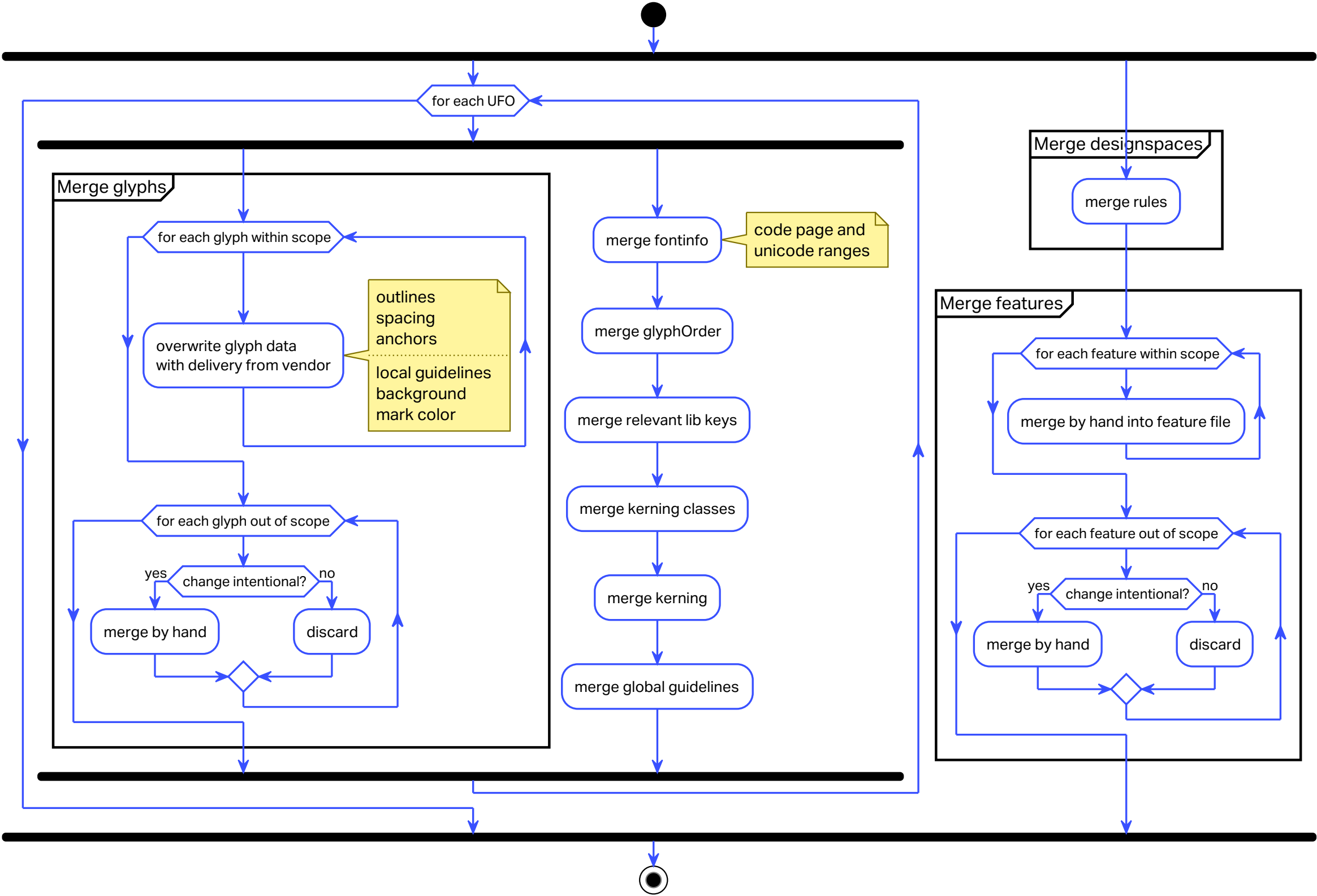
Google Sans Build Process – New Merge Process – Glyphs Vendor



Merge Process

Goal: avoid unintended changes,
keep intentional ones

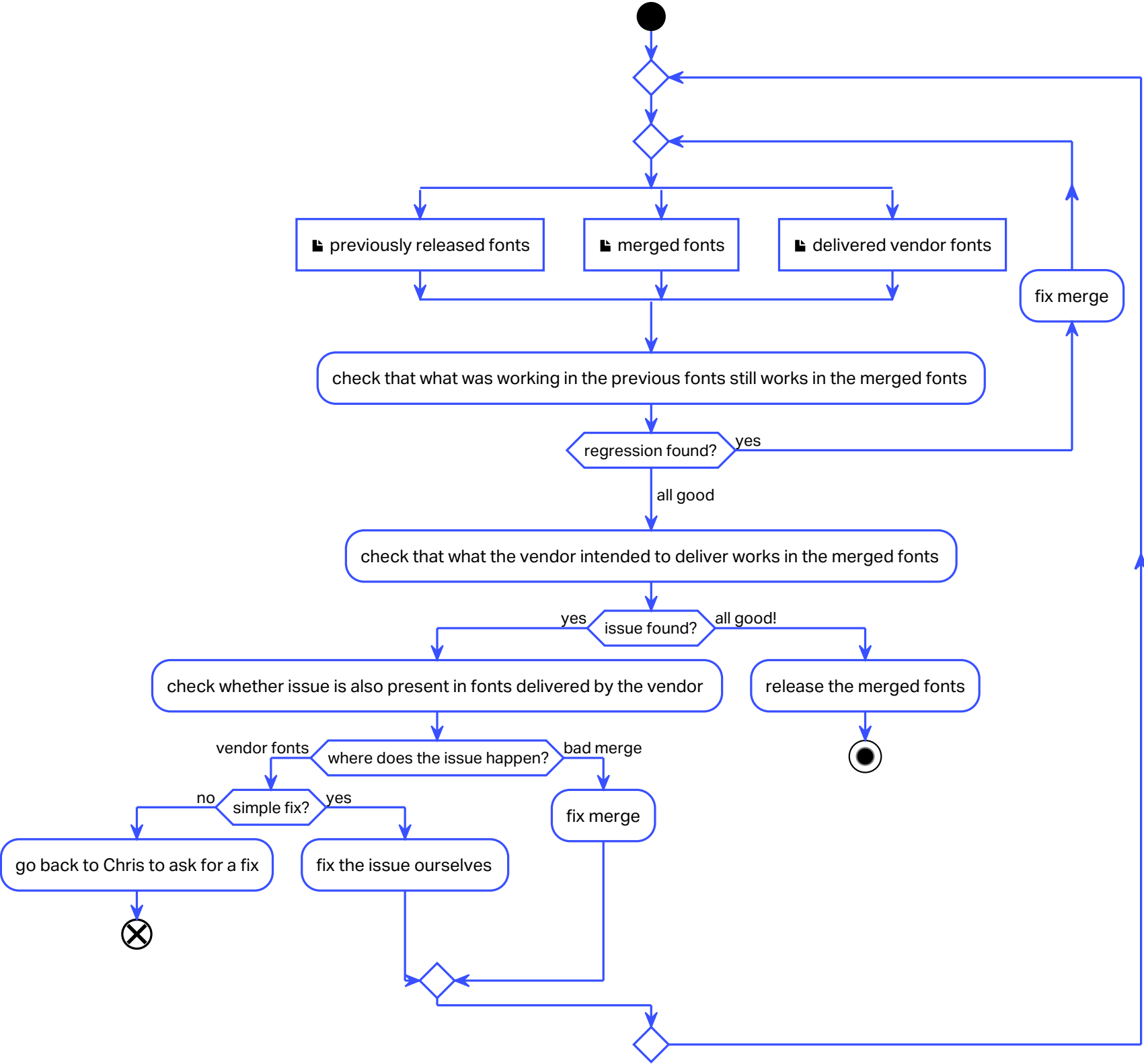
- Noise from the UFO-Glyphs conversion
- Glyph changes
 - Unintentional
 - Run RMX tools on the whole font instead of just a small scope
 - Glyphs.app moves components around sometimes
 - Non-functional changes such as changing mark colors, moving guidelines, changing background outlines...
 - Intentional
 - Add important guidelines for the script
 - Tweak a shape out of scope to uniformize between scripts
 - Take advantage of components across scripts
- Feature changes
 - Glyphs.app can regenerate some features, can be good (added glyphs to handle) or unintended
 - Manual adjustments to features (bug discovered in other script while testing in the scope)
 - Kerning out of scope glyphs as a consequence of class kerning mixing scripts



Quality Assurance Process

Ensure that the merge process preserves:

- existing functionality
- intended new functionality from vendors



Next Steps

Next Steps

- Turn our exploratory work into a working font building pipeline
- Document our merge process and delivery requirements for vendors
- Test our merge process with a work-in-progress delivery from one of the upcoming new scripts



Thank you

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